Pi & Planck

SCALE AND SPEED

How Fast Are You Moving?

Sitting still, you travel at four speeds at once. Each one is measured against something different.

SEEN FROM AFAR

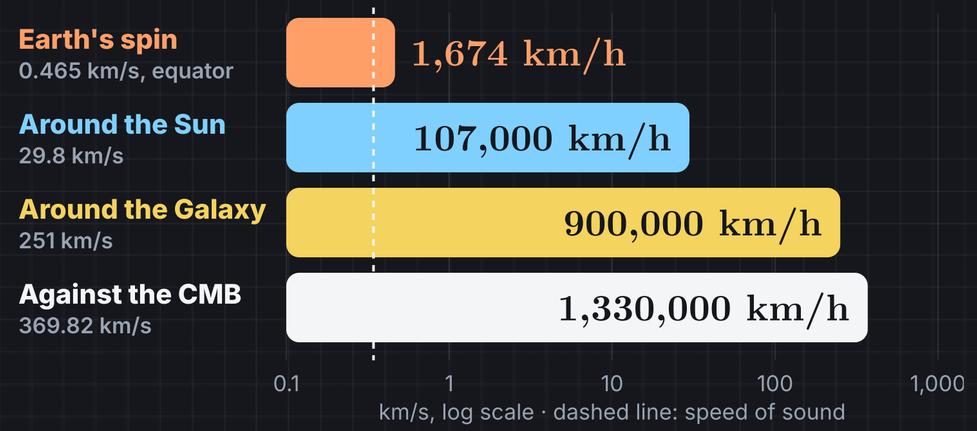


The Milky Way above Earth's glow

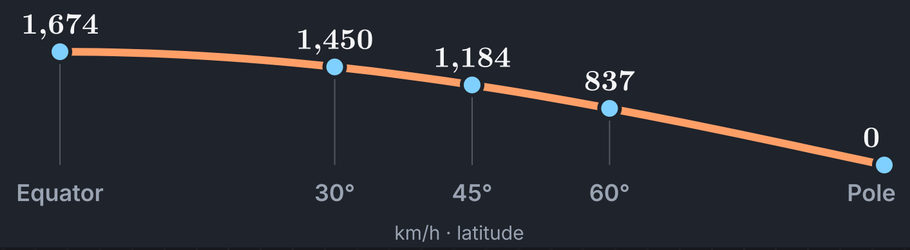


Earth, a speed in motion

FOUR SPEEDS AT ONCE



SPIN SPEED DROPS WITH LATITUDE



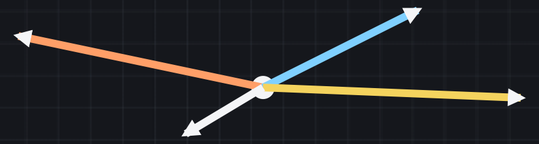
MEASURED AGAINST WHAT?

Earth's spin
the ground turning on Earth's axis

Around the Sun
Earth's path around the Sun

Around the Galaxy
the Sun's path around the Milky Way's centre

Against the CMB
the afterglow of the early universe



Do not simply add them: the directions differ.
One number each, never one total.

1,674 km/h — Earth's spin at the equator. At 45° latitude it is about 1,184 km/h; at the poles, 0.

29.8 km/s — Earth's mean speed around the Sun, about 107,000 km/h.

370 km/s — The Sun's speed against the cosmic microwave background (Planck: 369.82 ± 0.11 km/s).

Sources: NASA Earth Fact Sheet · NASA GSFC, The Rotating Earth · Zavada and Píška, Gaia DR3 kinematics, arXiv:2308.11060 (preprint) · Planck 2018 results I, A&A 641, A1 · Photos: NASA (ISS Expedition 73; Blue Marble 2012, NASA GSFC)